

Yes, your data is suitable for time-series forecasting.

You have columns like:

Category, City, Price, Day, Month, Year, Date

Your useful time-series fields are:

Date = time

Price = target value to predict

City = separate market/location

Category = price category/source

Your data range appears to be from 2024-05-05 to 2026-05-05, with daily records, so you have about 2 years of data. Good enough for a first daily/weekly/monthly forecasting model.

Steps

1. Clean the data

- Convert Date into proper date format.
- Convert Price into numeric format.
- Check missing dates, duplicate rows, and missing prices.
- Decide whether you want one model for each city or one combined model for all cities.

2. Choose forecasting level

- Daily forecast: use original daily data.
- Weekly forecast: group data by week and calculate average weekly price.
- Monthly forecast: group data by month and calculate average monthly price.

3. Start with one city first

- Example: Delhi, Chennai, Ahmedabad, etc.
- Do not begin with all cities together.
- Pick one city, build the model, visualize it, then repeat for other cities.

4. Visualize raw trend

- Plot Date on x-axis.
- Plot Price on y-axis.
- This shows whether price is increasing, decreasing, seasonal, or noisy.

5. Create train/test split

- Use older dates for training.
- Keep the most recent dates for testing.
- Example:
 - Train: first 80% of dates
 - Test: last 20% of dates

6. Try simple baseline first

- Use moving average.
- Use last known price.
- Use weekly/monthly average.
- This gives you a comparison point before using advanced models.

7. Use forecasting models

- a. For simple and explainable forecasting: ARIMA or SARIMA.
- b. For trend + seasonality + holidays: Prophet.
- c. For machine learning style: XGBoost, Random Forest, or LightGBM.
- d. For deep learning later: LSTM, but only after simpler models work.

8. Add useful features

- a. Day of week
- b. Month
- c. Week number
- d. Previous day price
- e. Previous 7-day average
- f. Previous 30-day average
- g. Price change from yesterday
- h. City name

9. Evaluate model

- a. Compare predicted price vs actual price.
- b. Use metrics like:
 - i. MAE: average error in price units
 - ii. RMSE: larger penalty for big mistakes
 - iii. MAPE: percentage error

10. Visualize predictions

- a. Line chart: actual price vs predicted price.
- b. Future forecast chart: next 7 days, next 4 weeks, next 3 months.
- c. Trend chart: upward/downward/stable.
- d. City comparison chart: compare prices across markets.

11. Create forecast outputs

- a. Daily forecast: next 7, 15, or 30 days.
- b. Weekly forecast: next 4, 8, or 12 weeks.
- c. Monthly forecast: next 3, 6, or 12 months.

12. Build a dashboard later

- a. City selector
- b. Forecast period selector: daily / weekly / monthly
- c. Actual vs predicted chart
- d. Future price chart
- e. Trend indicator: increasing, decreasing, stable
- f. Table of predicted prices

Best path for your project: start with **one city daily forecast**, then convert the same data into **weekly and monthly averages**, then build visuals. After that, expand to all cities.